

Christian Anfinsen explained denaturation of proteins by heat treatment leading to breakage of non-covalent bond.

Figure 8.24

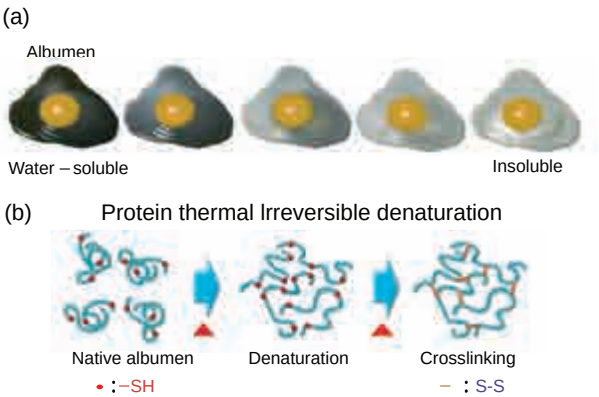


Figure 8.25: Protein denaturation

8.5.4 Protein Bonding

There are three types of chemical bonding

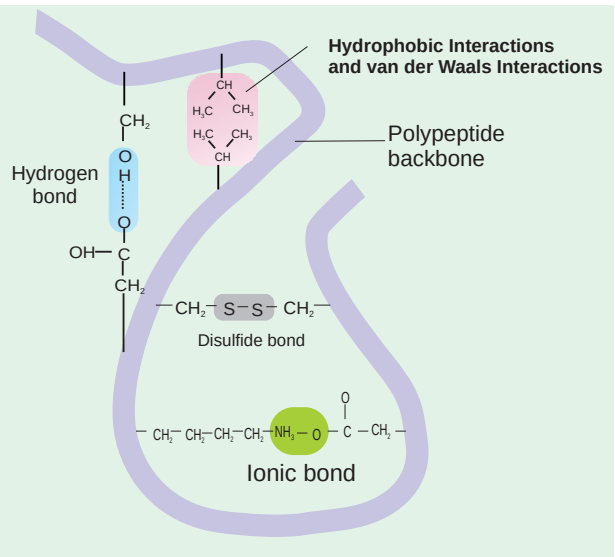


Figure 8.26: Protein bonding

Hydrogen Bond

It is formed between some hydrogen atoms of oxygen and nitrogen in polypeptide chain. The hydrogen atoms have a small positive charge and oxygen and nitrogen have small negative charge. Opposite charges attract to form hydrogen bonds.

Though these bonds are weak, large number of them maintains the molecule in 3D shape (Figure 8.26).

Ionic Bond

It is formed between any charged groups that are not joined together by peptide bond. It is stronger than hydrogen bond and can be broken by changes in pH and temperature.

Disulfide Bond

Some amino acids like cysteine and methionine have sulphur. These form disulphide bridge between sulphur atoms and amino acids.

Hydrophobic Bond

This bond helps some protein to maintain structure. When globular proteins are in solution, their hydrophobic groups point inwards away from water.



The more the distance between the sulphur atoms, the more the proteins bend; the more the hair curls.



8.5.5 Test for Proteins

The biuret test is used as an indicator of the presence of protein because it gives a purple colour in the presence of peptide bonds (–C–N–). To a protein solution an equal quantity of sodium hydroxide solution is added and mixed. Then a few drops of 0.5% copper (II) sulphate is added with gentle mixing. A distinct purple